Supplemental Material: Gradient concentration constrains trajectory diagnostics in variational quantum meta-learning

Sang Ho Oh[1]

[1] *Department of Computer Engineering and Artificial Intelligence, Pukyong National University, Busan 48513, Korea*

This Supplemental Material collects the supporting analyses referenced in the main text: the concentration of the identity prefactor (Sec. S1), the one-step versus multi-step identity check (Sec. S2), the association of alignment with the held-out loss rather than the gap (Sec. S3), the near-zero-mean-transfer character of the task family (Sec. S4), and four robustness checks—a term-type-stratified split, additional Hamiltonian families and entangler topologies, a finite-shot variance proxy, and a random initialization (Sec. S5); and a synthetic positive control showing that the joint control passes a genuine alignment–gap signal (Sec. S6). All quantities use the definitions, estimators, and hierarchical-bootstrap procedure of the main text. Rank correlations are Spearman coefficients; joint controls are partial rank correlations $\rho(A, G \mid I_{\mathrm{val}}, L_{\mathrm{val}}(\theta_0))$ unless stated otherwise. The underlying per-task tables are provided in the accompanying data repository.

S1. CONCENTRATION OF THE IDENTITY PREFACTOR

The first-order identity of the main text reads $I_{\mathrm{val}} \approx \alpha\, A\, q$ with prefactor $q = \|g_{\mathrm{tr}}\|\|g_{\mathrm{val}}\|$. The controlling role of gradient concentration is a statement about the relative spread of q, not of $\|g_{\mathrm{tr}}\|$ alone. Table SI reports the pooled coefficient of variation of $\|g_{\mathrm{tr}}\|$, $\|g_{\mathrm{val}}\|$, and q across tasks at each system size. Both gradient norms concentrate with scale, and the product q—being the product of two fluctuating norms—has a larger but likewise decreasing coefficient of variation. This is the regime in which the ranking of I_{val} collapses onto that of A.

TABLE SI. Pooled coefficient of variation of the two gradient norms and of the identity prefactor $q = \|g_{\mathrm{tr}}\|\|g_{\mathrm{val}}\|$, as a function of system size n (pooled over depth L and seeds).

n	CV($\|g_{\mathrm{tr}}\|$)	CV($\|g_{\mathrm{val}}\|$)	CV(q)
4	0.336	0.361	0.551
6	0.285	0.282	0.466
8	0.246	0.261	0.406
10	0.264	0.261	0.451

TABLE SII. Rank correlation of the support–validation alignment A with the validation improvement after one inner step ($I_{\mathrm{val}}^{(1)}$) and after $K = 10$ inner steps ($I_{\mathrm{val}}^{(K)}$), pooled over depth and seeds.

n	$\rho(A, I_{\mathrm{val}}^{(1)})$	$\rho(A, I_{\mathrm{val}}^{(K)})$
4	+0.95	+0.60
6	+0.96	+0.68
8	+0.97	+0.76
10	+0.97	+0.85

S2. ONE-STEP VERSUS MULTI-STEP IDENTITY

The identity is a first-order (tangent) expression; it is at its most accurate in the one-step limit. Table SII compares the rank correlation between the alignment A and the validation improvement measured after a single inner step ($I_{\mathrm{val}}^{(1)}$) with that measured after the full $K = 10$ inner steps ($I_{\mathrm{val}}^{(K)}$). At $K = 1$ the correlation is near-perfect and essentially flat in n (0.95–0.97), confirming that the tangent expression captures the one-step improvement directly. At $K = 10$ the correlation is lower and rises with scale (0.60 to 0.85): as gradients concentrate and adaptation steps shrink, the full trajectory moves closer to its initial tangent, and the identity increasingly predicts the multi-step improvement as well. The tightening reported in the main text is therefore the convergence of the K-step improvement toward the one-step tangent, not a change in the identity itself.

S3. ALIGNMENT VERSUS THE HELD-OUT LOSS

The main text controls the apparent association between A and the observable-generalization gap G. The same conclusion holds when the target is the held-out loss $L_{\mathrm{val}}(\theta_K)$ itself rather than the gap. Table SIII reports $\rho(A, L_{\mathrm{val}}(\theta_K))$ raw, controlling for the local improvement I_{val}, and controlling jointly for I_{val} and the initial baseline $L_{\mathrm{val}}(\theta_0)$. The raw association is moderate and negative but is removed by the joint control at every size ($|\rho| \leq 0.10$), so the vanishing residual is not special to the signed gap.

We also report the baseline-corrected gap $\Delta G =$

$[L_{\mathrm{val}}(\theta_K) - L_{\mathrm{tr}}(\theta_K)] - [L_{\mathrm{val}}(\theta_0) - L_{\mathrm{tr}}(\theta_0)] = I_{\mathrm{tr}} - I_{\mathrm{val}}$, which removes the initial baseline at the level of the target. Its raw association with A is $\rho(A, \Delta G) = -0.48, -0.55, -0.53, -0.59$ at $n = 4, 6, 8, 10$; controlling for I_{val} alone removes it, $\rho(A, \Delta G \,|\, I_{\mathrm{val}}) = +0.03, -0.03, -0.07, -0.10$.

S4. NEAR-ZERO-MEAN-TRANSFER REGIME

Table SIV quantifies the transfer content of the primary task family. Support adaptation improves the support loss substantially (mean I_{tr} from $+0.41$ to $+0.16$), but the mean validation improvement is statistically indistinguishable from zero at every size, with roughly half of tasks improving. The per-task spread of I_{val} remains nonzero and is the quantity the first-order identity predicts. This regime is a clean null-transfer setting by construction: coefficients are drawn independently and the linear observable loss over disjoint Pauli support and query subsets has no shared minimizer, so there is little mean observable transfer to detect.

S5. ROBUSTNESS CHECKS

The four checks below vary one ingredient of the setup at a time and confirm that the identity link and the vanishing joint-controlled gap association both persist.

A. Term-type-stratified support/query split

To rule out that the gap is an artifact of an unbalanced split, we stratify the support/query split so that XX, YY, ZZ, and Z terms are balanced across the two subsets, and compare against the uniform (random) split. Table SV shows that both splits preserve the high identity link and the near-zero joint-controlled gap association.

TABLE SIII. Alignment versus the held-out loss $L_{\mathrm{val}}(\theta_K)$: raw rank correlation, controlling for the local improvement I_{val}, and the joint control additionally removing the initial baseline $L_{\mathrm{val}}(\theta_0)$.

n	$\rho(A, L^K_{\mathrm{val}})$	$\rho(A, L^K_{\mathrm{val}} \,\|\, I_{\mathrm{val}})$	$\rho(A, L^K_{\mathrm{val}} \,\|\, I_{\mathrm{val}}, L^0_{\mathrm{val}})$
4	-0.56	-0.34	-0.06
6	-0.49	-0.39	-0.03
8	-0.45	-0.38	-0.10
10	-0.36	-0.38	-0.02

TABLE SIV. Transfer content of the primary task family: mean and standard deviation of the validation improvement I_{val}, fraction of tasks with $I_{\mathrm{val}} > 0$, and mean support improvement I_{tr}.

n	$\langle I_{\mathrm{val}} \rangle$	$\mathrm{std}(I_{\mathrm{val}})$	$\mathrm{frac}(I_{\mathrm{val}} > 0)$	$\langle I_{\mathrm{tr}} \rangle$
4	-0.008	0.283	0.47	0.410
6	$+0.002$	0.161	0.51	0.313
8	-0.003	0.096	0.49	0.242
10	$+0.001$	0.060	0.50	0.158

B. Additional Hamiltonian families and topologies

We repeat the analysis at $n = 6$ for three Hamiltonian families—the anisotropic XYZ family of the main text, a transverse-field Ising model (TFIM), and a fixed random-local-Pauli family—each with a linear-chain and a ring entangler. Table SVI shows that the identity link stays high ($\rho(A, I_{\mathrm{val}}) = 0.49$–$0.91$) and the joint-controlled gap association stays near zero (-0.17 to $+0.13$) throughout, so the mechanism is not specific to the XYZ family or the linear chain.

C. Finite-shot variance proxy

The observable-variance proxy V_{tr} is defined with a finite shot budget. To confirm that finite sampling does not remove the effect, we compare V_{tr} estimated at 100 and 1000 shots against the exact ($N_{\mathrm{shot}} \to \infty$) value across tasks. The rank agreement is $\rho = +0.84$ at 100 shots and $\rho = +0.98$ at 1000 shots, with a small positive sampling bias (mean offset $+0.003$ and $+0.001$, respectively). Finite-shot statistics therefore add, rather than remove, sources of the artifact.

D. Random initialization

Finally, we replace the Reptile meta-initialization with a plain random initialization ($\theta \sim \mathcal{N}(0, 0.1^2)$, no meta-

TABLE SV. Uniform versus term-type-stratified support/query split. The identity link $\rho(A, I_{\mathrm{val}})$ and the joint-controlled gap association $\rho(A, G \,|\, I_{\mathrm{val}}, L_{\mathrm{val}}(\theta_0))$ are preserved under stratification.

n	split	$\rho(A, I_{\mathrm{val}})$	$\rho(A, G \,\|\, I_{\mathrm{val}}, L^0_{\mathrm{val}})$
4	uniform	0.35	-0.29
4	stratified	0.50	$+0.10$
6	uniform	0.81	-0.03
6	stratified	0.72	-0.05
8	uniform	0.76	$+0.04$
8	stratified	0.86	-0.01

TABLE SVI. Identity link and joint-controlled gap association across three Hamiltonian families and two entangler topologies ($n = 6$, $L = 3$).

family	topology	$\rho(A, I_{\mathrm{val}})$	$\rho(A, G \mid I_{\mathrm{val}}, L^0_{\mathrm{val}})$
XYZ	linear	0.81	−0.03
XYZ	ring	0.89	+0.08
TFIM	linear	0.49	+0.13
TFIM	ring	0.77	+0.07
rand. Pauli	linear	0.78	−0.03
rand. Pauli	ring	0.91	−0.17

training) at $n = 6$, $L = 3$. The identity link is $\rho(A, I_{\mathrm{val}}) = +0.74$, the raw gap association is $\rho(A, G) = -0.27$, and it collapses under the joint control to $+0.07$—matching the Reptile arm (0.68, −0.42, −0.01). The identity and the artifact are thus properties of the first-order variational geometry rather than of the meta-initialization.

S6. SYNTHETIC POSITIVE CONTROL: THE PROTOCOL PASSES GENUINE SIGNAL

Every demonstration in the main text is a collapse case, so one may ask whether the joint control is simply too aggressive—whether it would erase a genuine alignment–gap association if one existed. A closed-form synthetic ensemble answers this. Per task we draw an alignment A, a norm product q with CV ≈ 0.45 (matching the experiments), the identity-channel improvement $I_{\mathrm{val}} = \alpha A q + \varepsilon$, and an independent baseline $L_{\mathrm{val}}(\theta_0)$; the gap is then constructed in two ways. In the *channels-only* construction, $G = -0.8\, I_{\mathrm{val}} + 0.8\, L_{\mathrm{val}}(\theta_0) + \varepsilon'$, so any A–G association flows entirely through the two controlled channels. In the *genuine-channel* construction a direct term $-0.5\, A$ is added, representing a real alignment→gap effect beyond local improvement and baseline. Three seeds of 150 tasks each give Table SVII: the channels-only association collapses under the joint control ($-0.65 \to -0.06$), exactly as in the physical experiments, while the genuine-channel association survives it ($-0.90 \to -0.62$; per-seed range $[-0.66, -0.58]$). The numerical values depend on the injected channel strength: sweeping the direct coefficient from 0.1 to 0.8 (channel noise s.d. 0.10) gives surviving joint-controlled correlations of −0.21, −0.33, −0.45, −0.62, and −0.77, so survival is robust down to injected strengths comparable to the channel noise. The controls of the falsification protocol therefore collapse channel artifacts without erasing genuine signal: a diagnostic that survives them is a real candidate, not a false negative.

TABLE SVII. Synthetic positive control. Rank correlation of alignment with the gap, raw and under the joint control, when the gap contains only the local-improvement and baseline channels versus when a genuine direct alignment channel is added (3 seeds $\times 150$ tasks, pooled).

construction	$\rho(A, G)$	$\rho(A, G \mid I_{\mathrm{val}})$	$\rho(A, G \mid I_{\mathrm{val}}, L^0_{\mathrm{val}})$
channels only	-0.65	-0.01	-0.06
genuine channel	-0.90	-0.45	-0.62